

CODE OF LIVE PULSE DETECTOR

```
#include <Wire.h >

#include <LiquidCrystal_I2C.h >

#include <SoftwareSerial.h >


#define heartbeatSensor A0

#define buzzer 4

#define safetyInputSwitch 5

int step=0;

unsigned long currentTime=0,previousTime=0;

int timeDifference=0;

int hb=0, previousHb=0;

int threshold=515;

int safetyCount=0;

SoftwareSerial Sim800L(2, 3);

LiquidCrystal_I2C lcd(0x27,16,2);

void setup()

{

pinMode(13,OUTPUT);

pinMode(heartbeatSensor,INPUT);

lcd.init();

lcd.backlight();

lcd.print("Pulse Reader");

delay(2000);

Sim800L.begin(9600);

pinMode(buzzer,OUTPUT);

pinMode(safetyInputSwitch,INPUT);

}

void loop()

{

readPulse();
```

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delay(20);
}

void sendSMS(String sms)
{
  Sim800L.println("AT");
  delay(100);
  Sim800L.println("AT+CMGS="+919872268895 "");
  delay(100);
  Sim800L.print(sms);
  Sim800L.write('26');
}

void readPulse()
{
  if(analogRead(heartbeatSensor) > threshold)
  {
    digitalWrite(13,HIGH);
    if(step!=2)
    {
      step=1;
    }
    if(step==1)
    {
      step=2;
      currentTime=millis();
      timeDifference=currentTime-previousTime;
      previousTime=currentTime;
      hb=60000/timeDifference;
    }
  }
  else

```

```

{
digitalWrite(13,LOW);
step=0;
}

if(hb <200 && hb >10)
{
if(previousHb!=hb)
{
previousHb=hb;
if(digitalRead(safetyInputSwitch)==HIGH)
{
lcd.clear();
lcd.print("Pulse: ");
lcd.print(hb);
lcd.print("/min");
lcd.setCursor(0,1);
if(hb>100)
{
safetyCount++;
lcd.print("High Pulse!!");
if(safetyCount==3)
{
String m="Patient is having ";
m+=hb;

m+=" pulse rate!!. Please check on the patient immediately and seek medical advice if the condition
continues.";

digitalWrite(buzzer,HIGH);
sendSMS(m);
delay(2000);
lcd.clear();
lcd.print("Take Propranolol");

```

```

lcd.setCursor(0,1);
lcd.print("1 tablet.");
delay(5000);
}
}
else if(hb <50)
{
safetyCount++;
lcd.print("Low Pulse!!");
if(safetyCount==3)
{
String m="Patient is having ";
m+=hb;

m+=" pulse rate!!. Immediate attention is advised. Please check the patient and consult a healthcare
professional.";

digitalWrite(buzzer,HIGH);
sendSMS(m);
delay(2000);
lcd.clear();
lcd.print("Take Atropine");
lcd.setCursor(0,1);
lcd.print("1 tablet.");
delay(5000);
}
}
else
{
lcd.print("Normal Rate");
digitalWrite(buzzer,LOW);
safetyCount=0;
}
}

```

```
}  
else  
{  
  lcd.clear();  
  lcd.print("Turn On");  
  lcd.setCursor(0, 1);  
  lcd.print(hb);  
}  
}  
}  
}
```